



Syrian Private University · Faculty of Artificial Intelligence Engineering

Enhanced Authentication in IoT

using Federated Learning

ScLBS Scalable · Self-Certified · FL-based



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Objective & Scientific Problem



Scientific Problem

- **Heavy classic auth** — Password / RSA / PKI → high compute, storage, comms on constrained devices
- **Central authority risk** — Certificate authorities → single point of failure, weak scalability
- **Dynamic membership** — Group-key schemes → rekeying overhead explodes on join / leave
- **Weak proofs** — Most schemes → informal analysis, no formal security verification



Research Objective

Design ScLBS — one integrated, self-certified authentication scheme for distributed IoT



Self-certified keys — no CA, no exposed credentials



FL-driven adaptive trust evaluation



Location-aware + zero-knowledge two-factor



B+-tree group keys → $O(\log n)$ rekeying



Formally proven · Dolev–Yao · ProVerif

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Key References

What each reference delivers → and how it maps to ScLBS

1	EAP-based Mutual Authentication (WLAN IoT)	Done: Lightweight EAP + symmetric keys; −99% compute, −98% energy; NS-3 + prototype	→ ScLBS: Shared NS-3 evaluation · forward secrecy · Dolev–Yao model
2	Auth & Defense Strategies for IoT Risks	Done: Multi-layer defense: OTP, ECC, anomaly detection, edge/cloud scaling	→ ScLBS: Backs EECC choice · scalability · privacy-by-design
3	A Survey of IoT Authentication Schemes	Done: 3-/5-layer model; per-layer attacks; protocol taxonomy	→ ScLBS: Threat baseline (Mirai, DoS, Sybil) · classification basis
4	P2P-based Authentication Protocol	Done: 2FA over Chord/Scramble; CoAP; response ≈ 0.016 s	→ ScLBS: Confirms 2FA · decentralization · ECC direction
5	Systematic Review & Security Assessment (2014–2023)	Done: Formal + informal assessment; 4 benchmarks	→ ScLBS: Defines metrics: compute · comms · energy · memory

Theoretical Background



Federated Learning (FL)

Collaborative trust learning — model updates, no raw data



Enhanced ECC (EECC)

320-bit curve · small keys · hardness of EECCLP



Self-Certified Keys / ZKP

RN issues public key — private key never leaves node



Location-Aware Auth

Geographic position as a second authentication factor



B+-tree Group Keys

Hierarchical rekeying in $O(\log_a n)$ on join / leave



Formal Verification

Dolev–Yao model · ProVerif · six security theorems

Justification — Environment Comparison

Criterion	Node-RED	NS-3	Cisco Packet Tracer
Interface	Visual GUI	Command-line	Visual GUI
Ease of use	Easy	Moderate	Easy
Flexibility	High	Very high	Low
Programming	Yes — Python crypto	Yes	No
Best suited for	Rapid prototyping & live demo	Academic perf. simulation	Learning & topology



Chosen: Node-RED — visual integration + Python crypto execution + live real-time demo

Suitability — Why Node-RED Fits



Flow-based on Node.js

wire devices · APIs · services visually



Browser visual editor

drag-and-drop GUI, low overhead



Runs Python crypto

EECC functions executed inside the flow



Live node management

sensing nodes orchestrated in real time



Real-time dashboard

analytics + debugging on live data

```

1: Input:  $\Omega_{SN}$ : public key of SN,  $P\Omega_{RN}$ : public key of RN,  $\mu_{SN}$ : SN's counter value,  $t_{SN}$ : position of SN,  $\mu_{RN}$ : RN's counter value.
2: Output: SK.
3: procedure Generation of SK
4: begin:
5:    $\mu_{SN} \leftarrow 0, \mu_{RN} \leftarrow 0$ ;
6:   for  $i \neq 1$  do
7:     while  $N \neq 1$  do
8:        $T_{SN} \leftarrow TR_{SN} + \Psi \pmod n$ ;
9:       SN generates a Session Request (SRQ) using  $ID_{SN}$  and  $T_{SN}$  and communicates it to RN;
10:    end while
11:    while  $M \neq 1$  do
12:       $T_{RN} \leftarrow TR_{RN} + \Psi \pmod n$ ;
13:       $ST_{RN} \leftarrow \Omega_{RN} + \dot{h}(ID_{SN} || t_{SN}) + \Psi + [\Gamma(\Omega_{SN}) + \dot{h}(ID_{SN})] \pmod n + \Omega_{SN}$ ;
14:       $SK \leftarrow TR_{RN} \times ST_{RN} + \chi_{RN} + T_{SN}$ 
15:       $SK \leftarrow (TR_{RN} \times ST_{RN} \pmod n) + \Psi + (\chi_{RN} \times SR_{RN} \pmod n) + \Psi$ ;
16:    end while
17:     $ST_{SN} \leftarrow \Omega_{SN} + \dot{h}(ID_{SN} || t_{SN}) + \Psi + [\Gamma(\Omega_{RN}) + \dot{h}(ID_{SN})] \pmod n + \Omega_{RN}$ ;
18:     $SK \leftarrow TR_{SN} \times ST_{SN} + \chi_{SN} + T_{RN}$ 
19:     $SK \leftarrow (TR_{SN} \times \chi_{RN} \pmod n) \times \Psi + (\chi_{RN} + SR_{SN} \pmod n) + \Psi$ ;
20:  end for
21:  Return SK;
22: end procedure
  
```

Algorithm 2. Pseudo code for generation of SK using FL.

Node-RED · live flow editor with embedded EECC crypto nodes

ScLBS — the proposed scheme

Self-certified · FL-driven · location-aware two-factor authentication built on EECC + hierarchical keys



01 · FL Self-Certified Framework

Distributed public-key derivation — no central certificate authority



02 · Two-Factor Authentication

Device identity + geographic location, reinforced by FL trust



03 · Zero-Knowledge Verification

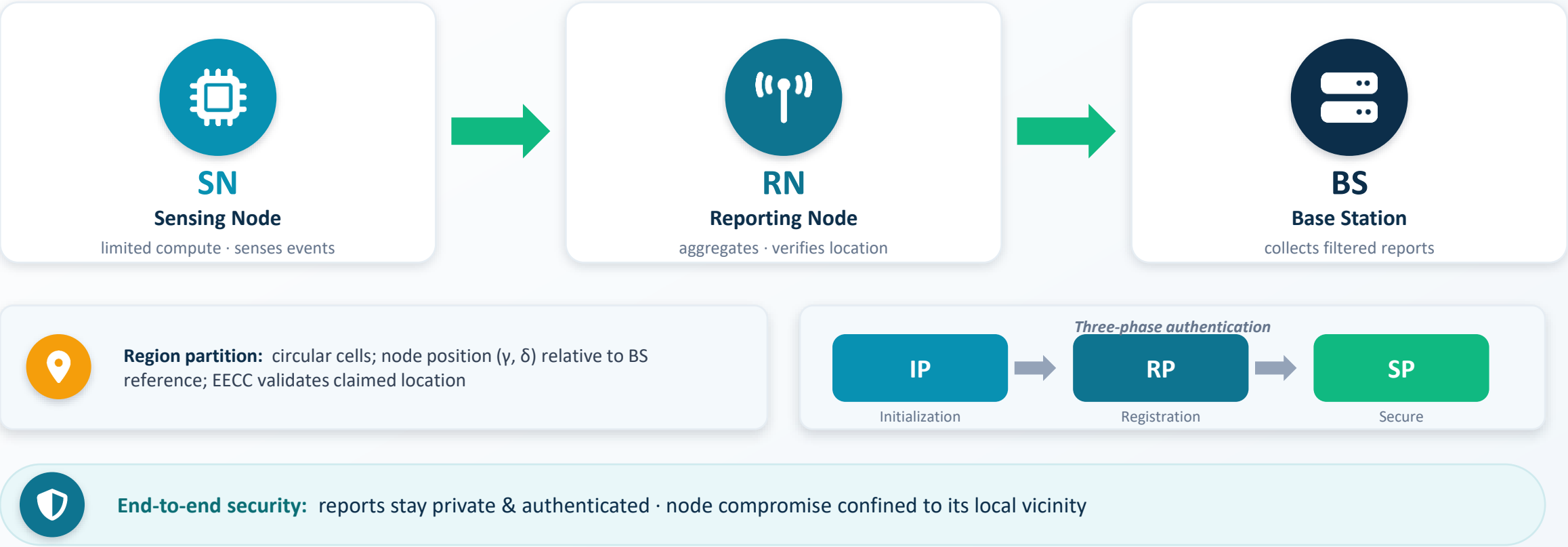
Verify public keys while private credentials stay hidden



04 · Hierarchical Session Keys

B+-tree keying ↓ latency · ↓ overhead · ↓ energy at scale

System Model & Overview



Initialization (IP)



Fix public parameters shared by all parties



Elliptic curve ξ over Φp field length $p = 320$ bits



Base point Ψ of order n



Shared secrets one-way hash $h(\cdot)$ + master key ΘSN

PUBLIC-KEY GENERATION

$$\Omega = \chi \times \Psi$$

EECC point multiplication · Eq. (1)

 Ω

public key

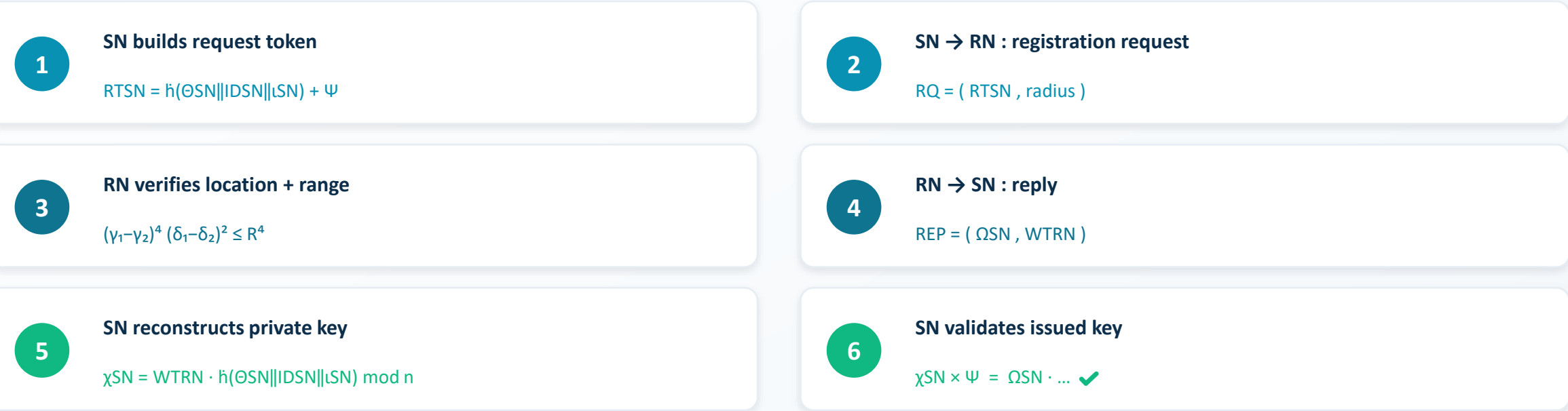
 χ

private key

 Ψ

base point

Registration (RP) — Self-Certified





Zero-knowledge · self-certified: RN issues & verifies the public key while the private key never leaves the SN

Trust Factor & Location Verification



Trust factor $\omega \in [0,1]$
initialized at 0.5, updated each time slot



Update rule
 $\omega = \omega \cdot w + \omega$ over slots $\tau_1 \dots \tau_n$

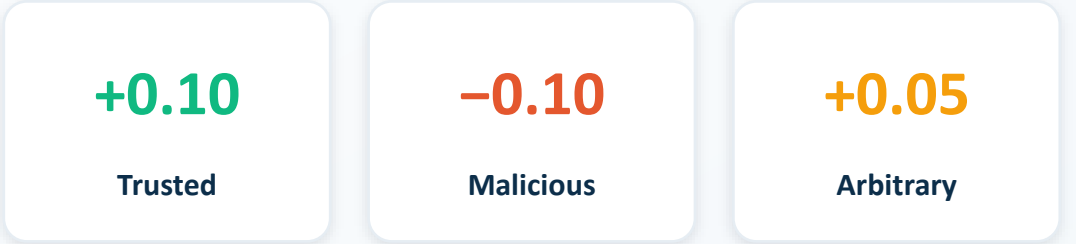


$\omega = 0 \rightarrow$ compromised
node removed from neighbour list

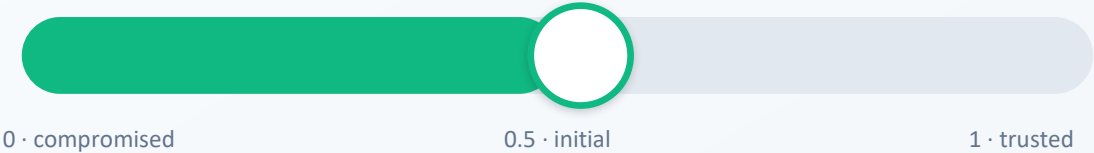


Max(ω_i) neighbour
most-trusted peer verifies the claimed location

BEHAVIOUR WEIGHTS (w)

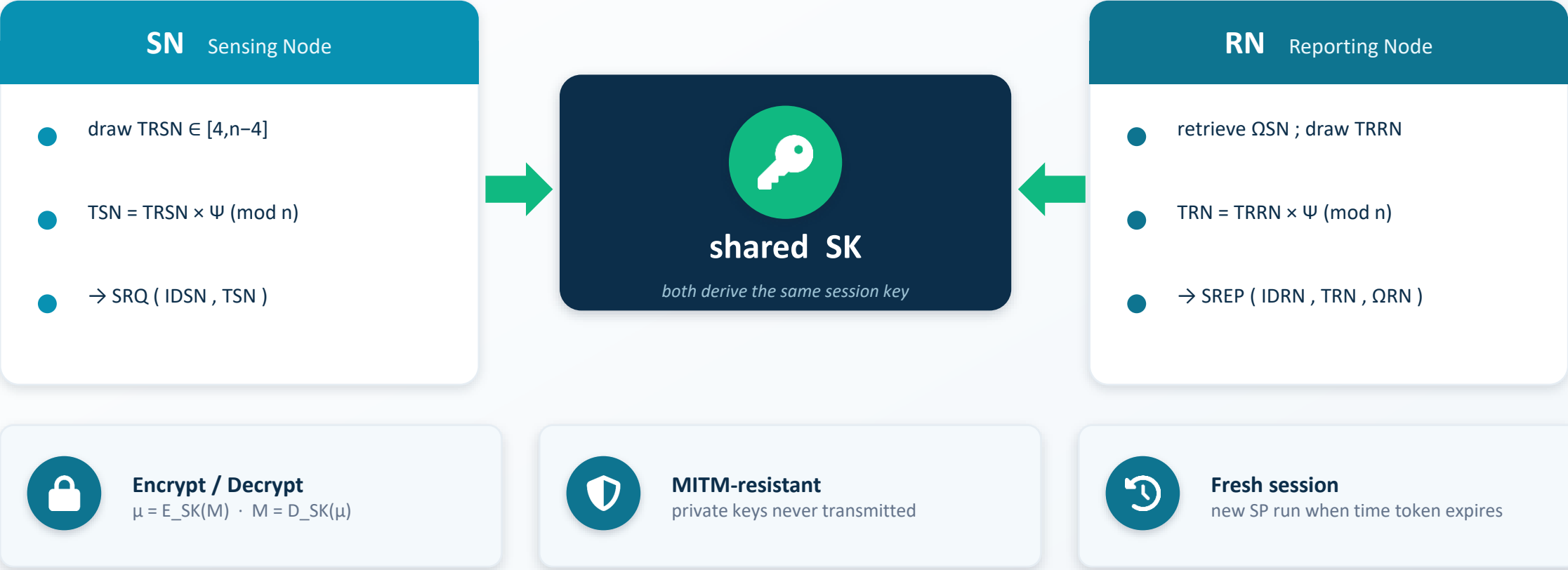


TRUST SCALE (ω)

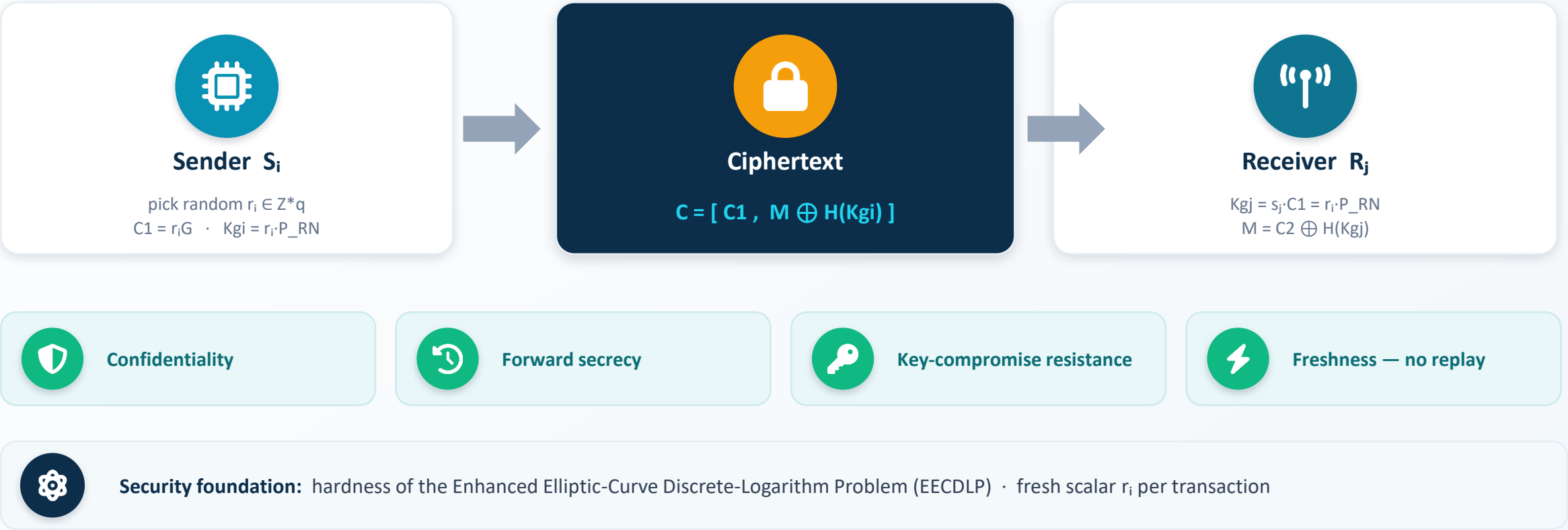


FL contribution: trust models refined collaboratively across the network — no raw data shared

Secure Phase (SP) — Session Key



Encryption & Decryption (EECC)



Hierarchical Group-Key Management



B+-tree organization
logarithmic $O(\log n)$ operations



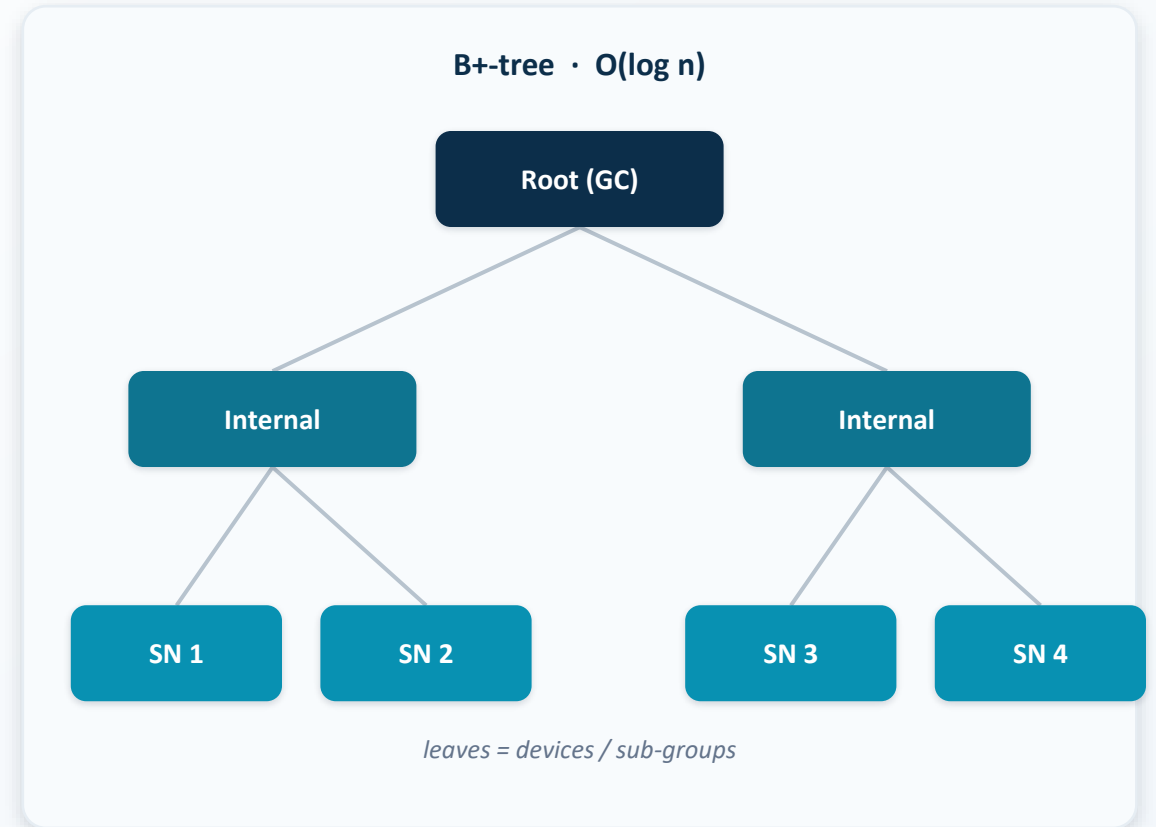
Localized rekeying
recomputed only on join / leave events



Sub-groups at leaves
split at threshold $d = m/2$, then rebalance



Group Controller (GC)
subtree rebalancing + selective key regen



Analysis & Formal Verification

SIX PROVEN THEOREMS

T1

Session-key secrecy

T2

Sybil resistance

T3

Collusion resistance

T4

Correctness of auth

T5

Forward secrecy

T6

Backward secrecy

ATTACK → GUARANTEE

**Sybil / forged IDs** → bind identity + location + EECC key**Node collusion** → FL regional trust confines impact**Key compromise** → forward secrecy — ephemeral keys**Post-revocation access** → B+-tree rekeying — backward secrecy**Replay / impersonation** → time-variant counters + zero-knowledge**Formally verified:** ProVerif under the Dolev–Yao adversary model — secrecy, mutual authentication, forward & backward secrecy all hold ✓

Demo — Node-RED

- 

Integration platform
flows orchestrate the full scheme
- 

Python crypto in-flow
EECC key & session operations
- 

Live node management
sensing nodes monitored in real time
- 

Dashboard + debug
real-time analytics on live traffic

```
(* Public channel *)
free c : channel.

(* Cryptographic primitives *)
fun h(bitstring) : bitstring.
fun ecMul(bitstring, bitstring) : bitstring.

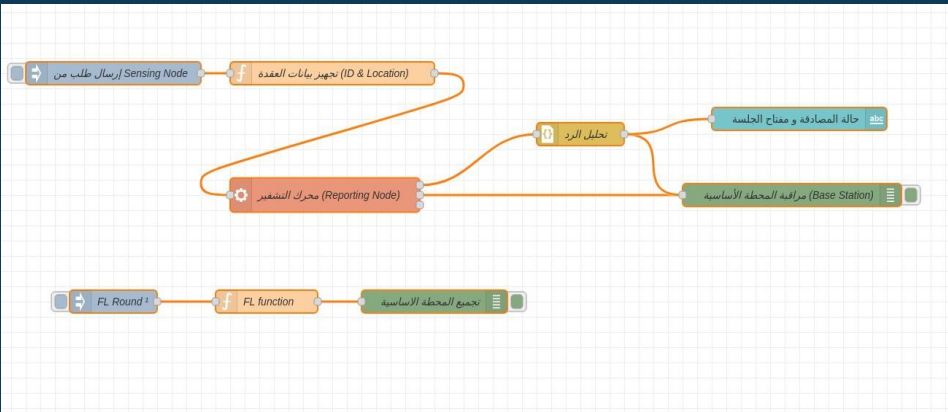
(* Secrets *)
free chiSN : bitstring [private].
free chiRN : bitstring [private].

(* SK *)
free SK : bitstring [private].

(* Authentication events *)
event beginAuth(bitstring).
event endAuth(bitstring).

process
(
  (* SN *)
  new rSN : bitstring;
  let OmegaSN = ecMul(chiSN, rSN) in
  event beginAuth(OmegaSN);
  out(c, OmegaSN);
  in(c, x : bitstring);
  let SK_SN = h(OmegaSN) in
  event endAuth(SK_SN)
)
|
(
  (* RN *)
  in(c, y : bitstring);
  let SK_RN = h(y) in
  out(c, SK_RN)
)
```

Debug output



Live dashboard · sensor & auth telemetry

Success · SN_101

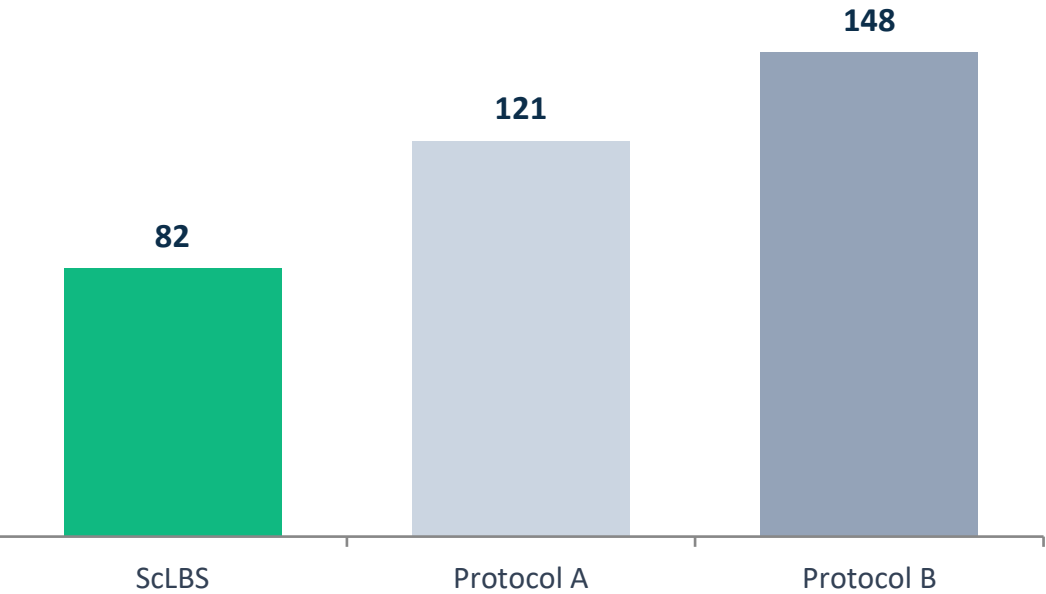
trust_score 0.94

FL round · completed

Results & Comparisons

NS-3 simulation · 20–100 devices · benchmarked vs. group-key protocols

Group-formation time (ms) · lower is better



–64%

network utilization vs. existing schemes



920 B

rekeying overhead · vs. 1,465–1,820 B



22.7 mJ

group-key energy · vs. 34.2–41.6 mJ

Ablation (latency ↑ if disabled): FL +17% · Location +12% · Hierarchy +23%

CONCLUSION

A balanced, provable authentication scheme for IoT



FL + self-certified keys

decentralized trust — no certificate authority



Location + zero-knowledge

compromise confined locally, private keys hidden



B+-tree key management

scalable rekeying in $O(\log d \ n)$



Proven + simulated

ProVerif (Dolev–Yao) + NS-3 evaluation



Outcome: lower authentication delay · reduced message overhead · improved network utilization · decreased EC computation cost

Where ScLBS goes next



Mobility-aware scenarios



Heterogeneous hardware



Real-world testbeds



Noisy / spoofed location robustness



Higher FL participation rates



Scale to smart-city size (10^3 – 10^4 nodes)



Current scope: results assume static nodes, ideal location verification, and homogeneous device capabilities